



QUANTITATIVE METHODS 1 · CHAPTER 5

Time Series

Detailed solutions

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1. Components and models

For each description identify trend, seasonality, cycle or irregular component: long-run e-commerce growth; December sales peaks; recession-related movements; a one-day system failure. When is a multiplicative model preferable?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. Long-run growth is trend; recurring December peaks are seasonality; recession movements are cyclical; the failure is irregular. A multiplicative model is preferable when seasonal amplitudes rise with the series level; an additive model suits roughly constant absolute seasonal effects.

2. Moving averages

Monthly demand for five months is 90, 105, 99, 120, 126. Calculate the three-period centred moving averages for months 2, 3 and 4.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Average each consecutive window of three observations and attach it to the middle month. Result. $MA_2 = (90 + 105 + 99) / 3 = 98$; $MA_3 = (105 + 99 + 120) / 3 = 108$; $MA_4 = (99 + 120 + 126) / 3 = 115$. Interpretation. Smoothing reveals an upward underlying movement while reducing month-to-month noise; endpoints are lost because a centred window requires neighbours.

3. Linear trend

For $t=1,2,3,4$ and $Y=(52,57,61,66)$ estimate $Y_t=a+bt$ using least squares and forecast $t=5$.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. $t=2.5$, $Y=59$. $\sum(t-t)(Y-Y)=23$ and $\sum(t-t)^2=5$, so $b=4.6$ and $a=59-4.6(2.5)=47.5$. Result. $Y_t=47.5+4.6t$ and $Y_5=70.5$. Interpretation. The fitted trend increases by 4.6 units per period; the forecast assumes no structural break.

4. Multiplicative seasonal forecast

Quarterly trend is $T_t = 100 + 3t$. Seasonal factors are .80, 1.05, .95 and 1.20 for Q1-Q4.

- 1 Check normalisation.
- 2 Forecast $t=9,10,11,12$.
- 3 Interpret the Q4 factor.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. The factors should average 1: $(.80 + 1.05 + .95 + 1.20)/4 = 1$. Compute T_t then multiply by the relevant seasonal factor. Result. Trends are 127, 130, 133 and 136. Forecasts are 101.6, 136.5, 126.35 and 163.2. Interpretation. Q4 is expected to be 20% above trend, not 20 percentage points above the previous quarter.

5. Forecast evaluation

Actual values are 100, 120, 110, 130. Model A forecasts 98, 118, 115, 128; Model B forecasts 105, 112, 108, 140. Calculate MAE for both and recommend a model. What else should be checked?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. Absolute errors for A are 2, 2, 5, 2, so $MAE_A = 2.75$. For B they are 5, 8, 2, 10, so $MAE_B = 6.25$. Model A performs better by MAE. Interpretation. The comparison should use a genuine validation period and also inspect bias, scale-sensitive versus percentage errors, residual patterns and performance after structural breaks.

6. Centred moving average and seasonal reading

Quarterly sales are 80, 100, 90, 130, 88, 110, 99 and 143.

- 1 Calculate the four-term moving averages.
- 2 Centre the moving averages for quarters 3 to 6.
- 3 Compute the observed/centred-moving-average ratios for quarters 3 to 6.
- 4 Explain what these ratios suggest.

Method prompt. A four-term average lies between quarters; centre it by averaging two consecutive four-term averages.

Detailed solution

Four-term moving averages. $MA1-4 = 100$; $MA2-5 = 102$; $MA3-6 = 104.5$; $MA4-7 = 106.75$; $MA5-8 = 110$.

Centred moving averages. Q3: $(100+102)/2 = 101$; Q4: $(102+104.5)/2 = 103.25$; Q5: $(104.5+106.75)/2 = 105.625$; Q6: $(106.75+110)/2 = 108.375$.

Ratios. Q3: $90/101 = 0.891$; Q4: $130/103.25 = 1.259$; Q5: $88/105.625 = 0.833$; Q6: $110/108.375 = 1.015$.

Interpretation. Q4 is well above the local trend and Q1 of the second year is below it. More years are required before estimating stable seasonal coefficients; four ratios alone are not sufficient.

7. Forecast validation and structural break

Actual monthly demand is 100, 104, 108, 150, 154 and 158. A trend model fitted on the first three months predicts 100, 104, 108, 112, 116 and 120.

- 1 Calculate forecast errors for months 4–6.
- 2 Calculate MAE and mean error over the validation period.
- 3 Diagnose the model.
- 4 Propose a responsible forecasting response.

Method prompt. Use error = actual – forecast. A persistently positive error signals systematic underforecasting.

Detailed solution

Errors. Month 4: $150 - 112 = 38$; month 5: $154 - 116 = 38$; month 6: $158 - 120 = 38$.

Validation metrics. $MAE = (38 + 38 + 38) / 3 = 38$. Mean error is also +38, showing systematic underforecasting rather than random noise.

Diagnosis. The level jumps between months 3 and 4 while the subsequent slope remains close to four units per month. This is consistent with a level structural break. A model trained only before the break is obsolete.

Response. Investigate the business event causing the jump, re-estimate the level using post-break data, compare with a naive benchmark, retain a validation window, and issue prediction intervals. Do not simply average pre- and post-break observations without explaining the regime change.